Trigger system

		> 10 mV/div	DC to 1 GHz	0.6 div
Trigger sensitivity (internal) [1]	2.5, 4, and 6 GHz bandwidth	≤ 10 mV/div	DC to 2 GHz	Greater of 1 div or 5 mVpp
			2.0 to 3.5 GHz	Greater of 1.5 div or 5 mVpp
		> 10 mV/div	DC to 2 GHz	0.6 div
			2.0 to 3.5 GHz	1.0 div
Trigger sensitivity (external) [1]	± 1.6 V	40 mVpp DC to 100 MHz, 70 mVpp 100 to 200 MHz		
	± 8 V	200 mVpp DC to 100 MHz, 350 mVpp 100 to 200 MHz		
Trigger level range	Any channel	± 6 div from center screen		
	External	8-V range = ± 8 V; 1.6-V range = ± 1.6 V		

1. Denotes warranted specifications; All others are typical. Specifications are valid after a 30-minute warm-up period and ± 10 °C from firmware calibration temperature.

Trigger type selections

Zone (hardware zone qualifier)	Trigger on user-defined zones drawn on the display. Applies to one analog channel at a time. Specify zones as either “must intersect” or “must not intersect.” Up to two zones. > 160,000 wfm/sec update rate Supported modes: Normal, peak detect, high resolution, max update rate Also works simultaneously with the serial decodes and mask limit test
Edge	Trigger on a rising and falling edge of any source, alternating or either edge of analog and digital channels
Edge then edge (B trigger)	Arm on a selected edge, wait a specified time, then trigger on a specified count of another selected edge. Minimum 4 ns
Pulse width	Trigger on a pulse on a selected channel, whose time duration is less than a value, greater than a value, or inside a time range Minimum duration setting: 2 ns Maximum duration setting: 10 s Range minimum: 10 ns
Pattern	Trigger when a specified pattern of high, low, and don’t-care levels on any combination of analog, digital, or trigger channels is [entered | exited]. Pattern must have stabilized for a minimum of 2 ns to qualify as a valid trigger condition Minimum duration setting: 2 ns Maximum duration setting: 10 s
Or	Trigger on any selected edges from available sources (analog and digital channels only). Bandwidth is 500 MHz
Rise/fall time	Trigger on rise time or fall time edge speed violations (< or >) based on user-selectable threshold. Select from (< or >) and time settings range between Minimum: 1 ns Maximum: 10 s
Nth edge burst	Trigger on the nth (1 to 65535) edge of a pulse burst. Specify idle time (10 ns to 10 s) for framing
Runt	Trigger on a position runt pulse that fails to exceed a high-level threshold. Trigger on a negative runt pulse that fails to drop below a low-level threshold. Trigger on either polarity runt pulse based on two threshold settings. Runt triggering can also be time-qualified (< or >) with a minimum time setting of 2 ns
Setup and hold	Trigger on setup/hold violations. Setup time can be set from –7 s to 10 s. Hold time can be set from 0 s to 10 ns. Minimum window (setup time + hold time) must be 3 ns or greater
Video	Trigger on all lines or individual lines, odd/even or all fields from composite video, or broadcast standards (NTSC, PAL, SECAM, PAM-M)
Enhanced video (HDTV) (Option)	Trigger on lines and fields of enhanced and HDTV standards (480p/60, 567p/50, 720p/50, 720p/60, 1080p/24, 1080p/25, 1080p/30, 1080p/50, 1080p/60, 1080i/50, 1080i/60)
ARINC429 (Option)	Trigger and decode on ARINC429 data. Trigger on word start/stop, label, label + bits, label range, error conditions (parity, word, gap, word or gap, all), all bits (eye), all 0 bits, all 1 bits
CAN (Option)	Trigger on CAN (controller area network) version 2.0A,2.0B, and CAN-FD (Flexible Data-rate) signals. Trigger on the start of frame (SOF), the end of frame (EOF), data frame ID, data frame ID and data (non-FD), data frame ID and data (FD), remote frame ID, remote or data frame ID, error frame, acknowledge error, from error, stuff error, CRC error, spec error (ack or form or stuff or CRC), all errors, BRS Bit (FD), CRC delimiter bit (FD), ESI bit active (FD), ESI bit passive (FD), overload frame., message, message and signal (non-FD), message and signal (FD, first 8 bytes only)
FlexRay (Option)	Trigger on frame ID or specific error condition, along with cycle-base and repetition-cycle filtering. Can also trigger on specific events such as BSS, TSS, FES, and wake up

Trigger type selections

I²C (Option)	Trigger at a start/stop condition or user-defined frame with address and/or data values. Also trigger on missing acknowledge, address with no acq, restart, EEPROM read, and 10-bit write
I²S (Option)	Trigger on 2's complement data of audio left channel or right channel (=, ≠, <, >, > <, < >, increasing value, or decreasing value)
LIN (Option)	Trigger on LIN (Local Interconnect Network) sync break, sync frame ID, frame ID and data , parity error, or checksum error
MIL-STD1553 (Option)	Trigger on MIL-STD 1553 signals on data word start/stop, command/status start/stop, RTA, RTA + 11 bits, and error conditions (parity, sync, Manchester)
SPI (Option)	Trigger on SPI (Serial Peripheral Interface) data pattern during a specific framing period. Supports positive and negative Chip Select framing as well as clock Idle framing and user-specified number of bits per frame. Supports MOSI and MISO data
UART/RS232/422/485 (Option)	Trigger on Rx or Tx start bit, stop bit, data content, or parity error
USB (Option)	Trigger on start of packet (SOP), end of packet (EOP), suspend***, resume***, reset***, packets (token, data, handshake, or special), and errors (PID, CRC5, CRC16, glitch, bit stuff***, SE1***). Supports USB 2.0 low-speed, full- speed, and hi-speed implementations. (Hi-speed is supported on ≥ 1-GHz models only)
SENT (Option)	Trigger and decode on SENT bus. start of fast channel message, start of slow channel message, fast channel SC and data, slow channel message ID, slow channel message ID and data, tolerance violation, fast channel CRC error, slow channel CRC error, all CRC errors, pulse period error, successive sync pulses error (1/64)
User-definable Manchester/NRZ (Option)	Trigger on start of frame (SOF), bus value, and Manchester errors
CXPI (Option)	Trigger on the start of frame (SOF), the end of frame (EOF), PTYPE, frame ID, data and info frame ID, data and info frame ID (long frame), CRC field error, parity error, inter-byte space error, inter-frame space error, framing error, data length error, sample error, all errors, sleep frame, wakeup pulse
USB PD (Option)	Trigger on preamble, EDP, ordered sets, preamble errors, CRC errors, header content (control messages, data messages, extended messages and value in HEX)

1. Suspend, resume, reset, bit stuff error, and SE1 error are USB 2.0 low- and full-speed only.

Search, navigate, and lister

Type		Edge, pulse width, rise/fall, runt, frequency peak, serial bus 1, serial bus 2
Copy		Copy to trigger, copy from trigger
Frequency peak	Source	Math functions
	Max number of peaks	11
	Control	Threshold, excursion, results order (frequency or amplitude)
Result display		Event lister or navigation. Manual or autoscroll via navigation or touch event lister entry to jump to a specific event

Waveform measurements

DC vertical accuracy/cursors [2]		Single cursor accuracy: ± [DC vertical gain accuracy + DC vertical offset accuracy + 0.21% full scale] Dual cursor accuracy: ± [DC vertical gain accuracy + 0.42% full scale] [1]
Number of measurements		56 automatic measurements, maximum of 10 displayed at a time
Cursors		2 pairs of XY cursors Automatic measurement of positions, ΔX, 1/ΔX, ΔY, and ΔY/ΔX
Automatic measurements		Measurements continuously updated with statistics. Cursors track last selected measurement. Select up to 10 measurements from the list below:
	Snapshot	Makes a snapshot of 31 most popular measurements. Touchable target to populate the measurement side bar
	Vertical	Peak-to-peak, maximum, minimum, amplitude, top, base, overshoot, preshoot, average- N cycles, average-full screen, DC RMS-N cycles, DC RMS-full screen, AC RMS-N cycles, AC RMS-full screen (standard deviation), ratio-N cycles, ratio-full screen Y at X
	Time	Period, frequency, counter, + width, - width, burst width, + duty cycle, - duty cycle, bit rate, rise time, fall time, delay, phase, X at min Y, X at max Y time at edge